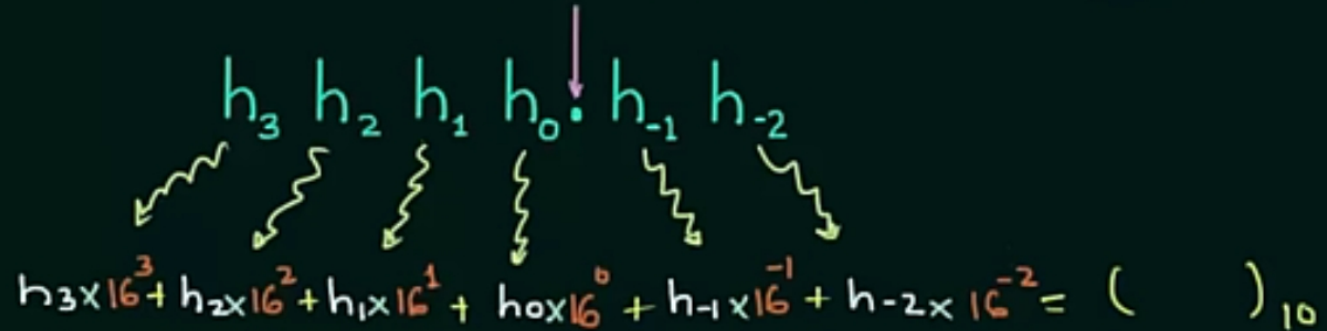


Hexadecimal to Decimal Conversion

Hexadecimal point

$$\gamma = 16$$


$$h_3 \times 16^3 + h_2 \times 16^2 + h_1 \times 16^1 + h_0 \times 16^0 + h_{-1} \times 16^{-1} + h_{-2} \times 16^{-2} = (\quad)_{10}$$



$$\text{Ex: - ?) } \begin{array}{ccc} 11 & 10 & 13 \\ \uparrow & \uparrow & \uparrow \\ (BAD)_{16} \end{array} \longrightarrow (29B9)_{10}$$

$$11 \times 16^2 + 10 \times 16^1 + 13 \times 16^0$$

$$11 \times 256 + 10 \times 16 + 13 \times 1$$

$$2816 + 160 + 13$$

$$2989$$

$$\begin{array}{l} 0 \\ \vdots \\ 9 \\ \underline{10} \rightarrow A \\ 11 \rightarrow B \\ \underline{\underline{12}} \rightarrow C \\ 13 \rightarrow D \\ 14 \rightarrow E \\ 15 \rightarrow F \end{array} \quad \begin{array}{l} 0 \text{ to } 15 \end{array}$$



ii) $(57.4)_{16} \rightarrow (87.25)_{10}$

$$5 \times 16^1 + 7 \times 16^0 + 4 \times 16^{-1}$$

$$80 + 7 + 0.25$$

$$(87.25)_{10}$$

$$4/16 \quad 1/4 = 0.25$$

H.W. $(BFF.C)_{16} \rightarrow (\quad)_{10}$

3071.75